# FULL STACK DEVELOPMENT – QUESTION PAPER

Based on Week 7 to Week 12

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| **Course** | Full Stack Development | **Course Code** | 20CS52I |
| **Maximum Marks** | 100 | **Time** | 3 Hours |

**Instructions:**

1. The question paper contains **5 sections**.
2. Each section contains **4 questions of 10 marks each** (40 marks printed in each section).
3. **Answer any TWO questions from each section.**
4. Each answered question carries **10 marks**.
5. Total marks to be answered = **100**.

## SECTION I – WEEK 7: SPRING AND SPRING BOOT

**1. (10 Marks)** Explain the Spring Framework, its need, advantages and important modules/components.

**2. (10 Marks)** Differentiate between Spring Framework and Spring Boot with respect to dependency injection, auto-configuration, server, boilerplate code and dependencies.

**3. (10 Marks)** Explain Dependency Injection in Spring. Describe Constructor Injection, Setter/Property Injection and Interface Injection with suitable examples.

**4. (10 Marks)** Explain Spring Beans, Autowiring, @Qualifier and different Spring Bean scopes.

## SECTION II – WEEK 8: API, REST, JDBC, ORM AND SPRING REST

**1. (10 Marks)** Explain Application Programming Interface (API). Describe how an API works, its need and the different types of APIs.

**2. (10 Marks)** Explain SOAP and REST API. Compare SOAP and REST with respect to data format, architecture, state, protocol and usage.

**3. (10 Marks)** Explain HTTP methods GET, POST, PUT and DELETE. Describe the request-response working of HTTP with a suitable REST API example.

**4. (10 Marks)** Explain Object Relational Mapping (ORM) and JPA. Discuss their features and benefits and explain the role of Spring RestController in creating RESTful web services.

## SECTION III – WEEK 9: NOSQL AND MONGODB

**1. (10 Marks)** Explain NoSQL databases. Discuss their features, benefits, schema-free nature, simple API and distributed characteristics.

**2. (10 Marks)** Explain the different types of NoSQL databases: Key-Value, Column-oriented, Graph-based and Document-oriented databases.

**3. (10 Marks)** Explain CAP theorem and BASE. Discuss Consistency, Availability and Partition

Tolerance with suitable examples.

4. **(10 Marks)** Explain MongoDB data types and MongoDB query operators. Discuss comparison, logical and array operators with suitable examples.

## SECTION IV – WEEK 10: SPRING REST, JPA AND DATABASE OPERATIONS

5. **(10 Marks)** Explain Spring REST Controller and the commonly used annotations such as @RestController, @GetMapping, @PostMapping, @PutMapping, @DeleteMapping and @RequestBody.

6. **(10 Marks)** Explain the important Spring/JPA annotations @Entity, @Id, @Autowired, @SpringBootApplication, @PathVariable and @RestController with their uses.

7. **(10 Marks)** Explain how to create a REST controller for CRUD operations. Describe the flow of request, controller, service/repository and database.

8. **(10 Marks)** Explain how a monolithic application can be converted into a microservices architecture. Describe the major steps involved in the migration.

## SECTION V – WEEK 11 & WEEK 12: TESTING, DEPLOYMENT, DOCKER AND KUBERNETES

1. **(10 Marks)** Explain Application Testing. Differentiate Manual Testing and Automation Testing and discuss Functional Testing, GUI Testing, Integration Testing, System Testing and Acceptance Testing.

2. **(10 Marks)** Explain the software deployment process and different deployment strategies such as Rolling, Blue-Green, Canary and A/B Testing.

3. **(10 Marks)** Explain Kubernetes and container orchestration. Discuss storage orchestration, automated rollouts/rollbacks, automatic bin packing, self-healing, and secret/configuration management.

4. **(10 Marks)** Explain Kubernetes Deployment and Service, and discuss Blue-Green and Canary deployment strategies. Also explain Kubernetes disaster recovery, backup and recovery, and the major types of disaster recovery.

**End of Question Paper**